

# ANAND BHALERAO

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## Experience

### FIS Global (formerly Dragonfly Financial Technologies, acquired 2024)

Pune, India

Software Engineer 2

Feb 2026 - Present

- Led real-time **session invalidation** post **\$2M fraud incident**, enabling instant threat containment; built atomic revocation (Spring + PostgreSQL) with zero downtime across multi-tenant systems.
- Architected centralized AES field-level encryption using a **custom MyBatis interceptor** for runtime SQL rewriting, **securing PII** across hundreds of query paths with **only 0.8ms overhead** over 300K+ records.
- Enabled indexed querying on encrypted data through **blind indexing**, reducing lookup latency by **over 700×**.
- Designed key rotation and **encrypted data migration pipelines** to support compliance and future cryptographic agility.
- Owned backend delivery of **FIDO2/WebAuthn biometric login**, implementing failure-threshold lockouts and creating a Jetpack Compose test harness that unblocked end-to-end integration with the native app team.

Software Engineer 1

Jan 2025 - Jan 2026

- **Owned end-to-end Zelle SSO integration** for 3 banking clients, collaborating with Zelle teams; **built 10+ Spring APIs** and 8 Angular interfaces, including lifecycle handling such as automated unenrollment on user deactivation.
- Reduced results per request from **90K to 10 rows**, by moving pagination, sorting, and filtering to the database layer, improving performance for **100K+ user** datasets.
- Eliminated **burst-traffic row-lock contention** caused by a single-row sequence table by migrating to **block-based cached sequence allocation**, improving throughput and reducing request timeouts.

Associate Software Engineer

Jul 2024 - Dec 2024

- Integrated **Outseer behavioral biometrics** into Angular authentication flows, collaborating with external teams to enable risk-based authentication using client-side signals.
- **Resolved frequent user lockouts** by enabling seamless switching between holding and subsidiary accounts; redesigned state handling to bypass redundant password checks for **multi-entity users**, improving overall user experience.

Software Development Engineer Intern | **Certificate**

Jan 2024 - Mar 2024

- Developed an end-to-end Electronic File Delivery feature, building **8 Angular screens** and **6 Spring REST APIs** to manage user preferences within a service-oriented architecture
- Engineered a homepage interstitial for mandatory contact updates, implementing admin-controlled retry logic to ensure high compliance.

## Projects

### EasyPay (UPI payment simulator)

Demo Video

Sarvatra Technologies (Sponsored Engineering Project)

- Architected a **distributed UPI payment** simulator modeled on NPCI switch architecture using Spring Boot, Redis, and RabbitMQ.
- Built fault-tolerant transaction processing with **idempotency**, concurrency control, and **Saga-based distributed transactions** with compensating actions, ensuring consistency across multi-service payment flows.
- Optimized UPI ID validation using **Redis-backed Bloom filters**, reducing database lookups by **92%** with controlled false-positive fallback.
- Instrumented **end-to-end observability** (OpenTelemetry + Prometheus) and **load tested up to 1,800 TPS**, achieving **85ms p95** and **140ms p99** latency under sustained load.

## Education

### Pune Institute of Computer Technology

Jun 2020 - Jun 2024

B.E in Computer Engineering (CGPA - 8.94)

Pune, Maharashtra

## Achievements

- Highest rating of **1763** on LeetCode; secured All India Ranks 145 and 496 in CodeChef Starters
- Solved **1100+** problems across LeetCode, CodeChef and Codeforces.
- Finalist in **Smart India Hackathon '22**.

## Technical Skills

**Languages:** Java, Kotlin, Typescript

**Technologies:** Spring Boot, MyBatis, JPA, Hibernate, Redis, Angular, Jetpack Compose, Compose Multiplatform

**Databases:** Oracle, PostgreSQL, MySQL